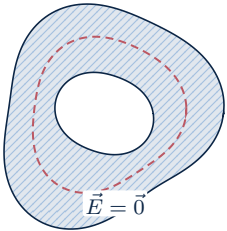
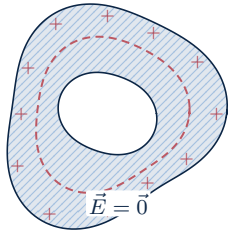
(a) Neutro, cavidad vacía



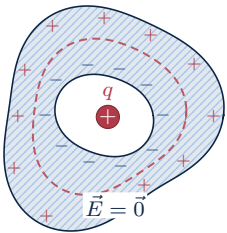
$Q_{\text{int}} = 0$
$Q_{\text{ext}} = 0$
$\vec{E}_{\text{cav}} = \vec{0}$
$\vec{E}_{\text{ext}} = \vec{0}$

(b) Carga Q, cavidad vacía



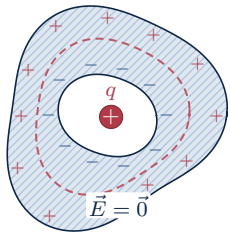
$Q_{\text{int}} = 0$
$Q_{\text{ext}} = Q$
$\vec{E}_{\text{cav}} = \vec{0}$
$\vec{E}_{\text{ext}} \neq \vec{0}$

(c) Neutro, carga q en la cavidad



$Q_{\text{int}} = -q$
$Q_{\text{ext}} = +q$
$\vec{E}_{\text{cav}} \neq \vec{0}$
$\vec{E}_{\text{ext}} \neq \vec{0}$

(d) Carga Q, carga q en la cavidad



$Q_{\text{int}} = -q$
$Q_{\text{ext}} = Q + q$
$\vec{E}_{\text{cav}} \neq \vec{0}$
$\vec{E}_{\text{ext}} \neq \vec{0}$

----- superficie de Gauss (cerrada) dentro del metal